

Brutal Practice Paper B27 - Motion & Time | Comparing Quantities | Heat | Simple Equations

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A cyclist rides 90 km out at 45 km/h, then the same 90 km back at 90 km/h. What is her average speed for the whole trip? [\[Motion & Time\]](#)
 - (A) 45 km/h
 - (B) 55 km/h
 - (C) 90 km/h
 - (D) 67.5 km/h
 - (E) 65 km/h
 - (F) 72 km/h
 - (G) 60 km/h

2. In a class of 40 students, 35% wear glasses, and of those, 4 also wear contact lenses. How many wear glasses but not contacts? [\[Comparing Quantities\]](#)
 - (A) 6
 - (B) 24
 - (C) 10
 - (D) 18
 - (E) 35
 - (F) 4
 - (G) 14

3. A metal block's temperature rises from 28 deg C to 92 deg C. What is the rise in temperature? [\[Heat\]](#)
 - (A) 64 deg C
 - (B) 56 deg C
 - (C) 32 deg C
 - (D) 70 deg C
 - (E) 92 deg C
 - (F) 60 deg C

4. Solve for x: $5x - 7 = 38$. [\[Simple Equations\]](#)
 - (A) 9
 - (B) 31
 - (C) 8
 - (D) 6.2
 - (E) 9.4
 - (F) 45
 - (G) 7.8

5. A train moves at 72 km/h and a bus at 25 m/s on parallel tracks. By how many m/s is the faster vehicle ahead in speed? [\[Motion & Time\]](#)
- (A) 47 m/s
 - (B) 25 m/s
 - (C) 52 m/s
 - (D) 10 m/s
 - (E) 2 m/s
 - (F) 20 m/s
 - (G) 5 m/s
6. Two numbers are in the ratio 3 : 5. What percent of the larger number is the smaller number? [\[Comparing Quantities\]](#)
- (A) 80%
 - (B) 166.7%
 - (C) 53.3%
 - (D) 40%
 - (E) 37.5%
 - (F) 60%
7. A clinical thermometer is marked from 35 deg C to 42 deg C. What is the size of this range? [\[Heat\]](#)
- (A) 35 deg C
 - (B) 8 deg C
 - (C) 6 deg C
 - (D) 7 deg C
 - (E) 42 deg C
 - (F) 77 deg C
8. Solve for x: $3(x + 4) = 27$. [\[Simple Equations\]](#)
- (A) 5.4
 - (B) 7
 - (C) 11
 - (D) 9
 - (E) 3
 - (F) 5
9. A pendulum completes 40 oscillations in 1 minute 20 seconds. What is its time period? [\[Motion & Time\]](#)
- (A) 0.5 s
 - (B) 0.67 s
 - (C) 3 s
 - (D) 2.5 s
 - (E) 1.5 s
 - (F) 40 s
 - (G) 2 s

- 10.** A jacket priced Rs 2000 gets a 20% discount, then a further 10% off the reduced price. What is the final price? [\[Comparing Quantities\]](#)
- (A) Rs 1600
 - (B) Rs 1440
 - (C) Rs 1400
 - (D) Rs 1444
 - (E) Rs 1500
 - (F) Rs 1480
 - (G) Rs 1320
- 11.** Midday temperatures on four days were 31 deg C, 35 deg C, 29 deg C and 33 deg C. What was the average? [\[Heat\]](#)
- (A) 33 deg C
 - (B) 31 deg C
 - (C) 34 deg C
 - (D) 128 deg C
 - (E) 32 deg C
 - (F) 30 deg C
- 12.** A number is tripled and then increased by 8 to give 50. Find the number. [\[Simple Equations\]](#)
- (A) 42
 - (B) 12.6
 - (C) 21
 - (D) 16
 - (E) 14
 - (F) 19.3
- 13.** A car travels for 2 hours 15 minutes at a steady 60 km/h. How far does it go? [\[Motion & Time\]](#)
- (A) 135 km
 - (B) 132 km
 - (C) 140 km
 - (D) 120 km
 - (E) 150 km
 - (F) 90 km
- 14.** A shopkeeper buys a fan for Rs 1200 and sells it for Rs 1500. What is the profit percent? [\[Comparing Quantities\]](#)
- (A) 25%
 - (B) 300%
 - (C) 30%
 - (D) 15%
 - (E) 12.5%
 - (F) 40%

- 15.** A thermometer has 10 equal small divisions between 30 deg C and 35 deg C. What does each small division represent? [\[Heat\]](#)
- (A) 0.05 deg C
 - (B) 5 deg C
 - (C) 0.5 deg C
 - (D) 0.2 deg C
 - (E) 1 deg C
 - (F) 10 deg C
- 16.** Solve for x: $x/4 + 3 = 11$. [\[Simple Equations\]](#)
- (A) 8
 - (B) 32
 - (C) 24
 - (D) 4
 - (E) 44
 - (F) 2
 - (G) 56
- 17.** A runner must cover 3 km at a steady 5 m/s. How long will it take, in minutes? [\[Motion & Time\]](#)
- (A) 10 min
 - (B) 12 min
 - (C) 20 min
 - (D) 15 min
 - (E) 6 min
 - (F) 600 min
- 18.** Find the simple interest on Rs 8000 at 5% per annum for 3 years. [\[Comparing Quantities\]](#)
- (A) Rs 2400
 - (B) Rs 240
 - (C) Rs 400
 - (D) Rs 9200
 - (E) Rs 1200
 - (F) Rs 1320
 - (G) Rs 1260
- 19.** Two bodies, at 80 deg C and 50 deg C, are placed in contact. In which direction does heat flow, and what temperature difference drives it? [\[Heat\]](#)
- (A) From 80 to 50; 130 deg C
 - (B) From 50 deg C to 80 deg C; 30 deg C
 - (C) Both ways; 65 deg C
 - (D) No flow; 0 deg C
 - (E) From 50 to 80; 130 deg C
 - (F) From 80 to 50; 15 deg C
 - (G) From 80 deg C to 50 deg C; difference 30 deg C

20. Solve for x : $2x + 5 = x + 19$. [\[Simple Equations\]](#)

- (A) 24
- (B) 12
- (C) 8
- (D) -14
- (E) 14
- (F) 4.6

21. A car covers the first 120 km in 3 hours and the next 120 km in 2 hours. What is its average speed?

[\[Motion & Time\]](#)

- (A) 44 km/h
- (B) 60 km/h
- (C) 50 km/h
- (D) 48 km/h
- (E) 40 km/h
- (F) 100 km/h

22. A town's population grew from 25000 to 30000 in one year. What is the percentage increase? [\[Comparing](#)

[Quantities\]](#)

- (A) 5%
- (B) 25%
- (C) 10%
- (D) 20%
- (E) 15%
- (F) 16.7%
- (G) 120%

23. A cup of tea cools from 90 deg C to 54 deg C in 12 minutes. What is the average drop in temperature per minute? [\[Heat\]](#)

- (A) 2 deg C/min
- (B) 6 deg C/min
- (C) 12 deg C/min
- (D) 3 deg C/min
- (E) 9 deg C/min
- (F) 4.5 deg C/min

24. Reena is 4 years older than twice her brother's age. If Reena is 26, how old is her brother? [\[Simple](#)

[Equations\]](#)

- (A) 7
- (B) 30
- (C) 15
- (D) 22
- (E) 11
- (F) 13
- (G) 9

25. A scooter's speed rises from 40 km/h to 50 km/h. What is the percentage increase in speed? [\[Motion & Time\]](#)
- (A) 20%
 - (B) 10%
 - (C) 2.5%
 - (D) 100%
 - (E) 50%
 - (F) 12.5%
 - (G) 25%
26. Rs 7200 is divided between A and B in the ratio 5 : 4. How much does B get? [\[Comparing Quantities\]](#)
- (A) Rs 800
 - (B) Rs 2400
 - (C) Rs 2880
 - (D) Rs 3200
 - (E) Rs 4500
 - (F) Rs 4000
 - (G) Rs 3600
27. Fifty rails, each 12 m long, are laid end to end with a 4 mm expansion gap after each rail. What is the total length of all the gaps? [\[Heat\]](#)
- (A) 20 cm
 - (B) 50 cm
 - (C) 0.2 cm
 - (D) 100 cm
 - (E) 200 cm
 - (F) 2 cm
 - (G) 20 mm
28. Solve for x: $7x - 4 = 3x + 20$. [\[Simple Equations\]](#)
- (A) 6
 - (B) 2
 - (C) 3
 - (D) 24
 - (E) -6
 - (F) 4
29. A car's odometer reads 45670 km at 8:00 am and 45910 km at 11:00 am. What was its average speed? [\[Motion & Time\]](#)
- (A) 90 km/h
 - (B) 240 km/h
 - (C) 45910 km/h
 - (D) 120 km/h
 - (E) 30 km/h
 - (F) 80 km/h
 - (G) 60 km/h

30. A 250 g solution contains 15% salt by mass. How many grams of salt does it contain? [\[Comparing Quantities\]](#)
- (A) 212.5 g
 - (B) 30 g
 - (C) 3.75 g
 - (D) 25 g
 - (E) 75 g
 - (F) 37.5 g
 - (G) 15 g
31. During the day, land heats faster than the sea. If the land is at 38 deg C and the sea surface at 29 deg C, what temperature difference drives the sea breeze? [\[Heat\]](#)
- (A) 10 deg C
 - (B) 67 deg C
 - (C) 38 deg C
 - (D) 8 deg C
 - (E) 29 deg C
 - (F) 19 deg C
 - (G) 9 deg C
32. Solve for x: $(2x - 1)/3 = 5$. [\[Simple Equations\]](#)
- (A) 2
 - (B) 12
 - (C) 7
 - (D) 9
 - (E) 23
 - (F) 8
33. A car goes 36 km in 30 minutes at a steady speed. Express this speed in m/s. [\[Motion & Time\]](#)
- (A) 72 m/s
 - (B) 10 m/s
 - (C) 36 m/s
 - (D) 12 m/s
 - (E) 25 m/s
 - (F) 18 m/s
 - (G) 20 m/s
34. After a 25% discount, a book costs Rs 270. What was its original price? [\[Comparing Quantities\]](#)
- (A) Rs 202.50
 - (B) Rs 345
 - (C) Rs 295
 - (D) Rs 360
 - (E) Rs 337.50
 - (F) Rs 540

- 35.** Two identical cans, one black and one white, start at 25 deg C in sunlight. After an hour the black is 49 deg C and the white is 37 deg C. How much more did the black can rise? [\[Heat\]](#)
- (A) 11 deg C
 - (B) 24 deg C
 - (C) 49 deg C
 - (D) 6 deg C
 - (E) 86 deg C
 - (F) 12 deg C
- 36.** A pen costs Rs 5 more than a pencil. Together they cost Rs 23. Find the cost of the pencil. [\[Simple Equations\]](#)
- (A) Rs 11.50
 - (B) Rs 4
 - (C) Rs 5
 - (D) Rs 9
 - (E) Rs 18
 - (F) Rs 13
 - (G) Rs 14
- 37.** Two friends start together and walk the same way, one at 5 km/h and the other at 7 km/h. How far apart are they after 90 minutes? [\[Motion & Time\]](#)
- (A) 4 km
 - (B) 3.5 km
 - (C) 18 km
 - (D) 10.5 km
 - (E) 3 km
 - (F) 1.5 km
- 38.** In an exam, 60% of the students passed. If there were 350 students, how many failed? [\[Comparing Quantities\]](#)
- (A) 60
 - (B) 140
 - (C) 160
 - (D) 290
 - (E) 40
 - (F) 210
 - (G) 175
- 39.** Which list arranges materials from BEST to WORST conductor of heat? [\[Heat\]](#)
- (A) Copper, wood, iron, water, air
 - (B) Iron, copper, wood, water, air
 - (C) Copper, iron, water, wood, air
 - (D) Copper, water, iron, air, wood
 - (E) Wood, copper, iron, air, water
 - (F) Air, wood, water, iron, copper
 - (G) Water, copper, iron, wood, air

40. Solve for x: $4(2x - 3) = 28$. [\[Simple Equations\]](#)

- (A) 4
- (B) 6
- (C) 5
- (D) 8
- (E) 5.5
- (F) 2.5

41. A bus covers 60 km at 30 km/h and another 60 km at 60 km/h. What is the average speed? [\[Motion & Time\]](#)

- (A) 45 km/h
- (B) 60 km/h
- (C) 50 km/h
- (D) 48 km/h
- (E) 40 km/h
- (F) 90 km/h
- (G) 30 km/h

42. Rs 5000 earns Rs 900 as simple interest in 3 years. What is the rate of interest per annum? [\[Comparing Quantities\]](#)

- (A) 9%
- (B) 6.5%
- (C) 3%
- (D) 5%
- (E) 6%
- (F) 18%
- (G) 60%

43. Four temperature readings have an average of 30 deg C. Three of them are 28 deg C, 31 deg C and 27 deg C. What is the fourth reading? [\[Heat\]](#)

- (A) 29 deg C
- (B) 28.7 deg C
- (C) 120 deg C
- (D) 34 deg C
- (E) 36 deg C
- (F) 30 deg C
- (G) 32 deg C

44. The sum of three consecutive integers is 72. What is the largest of them? [\[Simple Equations\]](#)

- (A) 72
- (B) 25
- (C) 27
- (D) 21
- (E) 26
- (F) 24
- (G) 23

45. A pendulum takes 90 seconds to complete 60 swings. Keeping the same length, how many swings will it make in 3 minutes? [\[Motion & Time\]](#)
- (A) 180
 - (B) 60
 - (C) 90
 - (D) 100
 - (E) 240
 - (F) 120
46. The ratio of boys to girls in a school is 7 : 5. If there are 24 more boys than girls, how many students are there in all? [\[Comparing Quantities\]](#)
- (A) 60
 - (B) 144
 - (C) 120
 - (D) 132
 - (E) 288
 - (F) 72
 - (G) 84
47. Why does a clinical thermometer have a kink (a narrow constriction) just above the bulb? [\[Heat\]](#)
- (A) It stops the mercury from flowing back, so the reading holds until the thermometer is shaken down
 - (B) It lets the thermometer measure higher temperatures
 - (C) It lets air escape from the tube
 - (D) It makes the mercury rise faster when heated
 - (E) It makes the scale easier to read at a distance
 - (F) It marks the normal body temperature
 - (G) It protects the bulb from breaking
48. Solve for x: $9 - 2x = 1$. [\[Simple Equations\]](#)
- (A) 2
 - (B) 3
 - (C) 4
 - (D) 8
 - (E) -4
 - (F) 5
 - (G) -5
49. Thunder is heard 5 seconds after the lightning is seen. If sound travels at 340 m/s, how far away was the lightning? [\[Motion & Time\]](#)
- (A) 68 m
 - (B) 3400 m
 - (C) 1700 m
 - (D) 340 m
 - (E) 68 km
 - (F) 170 m
 - (G) 850 m

50. 40% of the students play a sport, and 25% of those play cricket. What percent of ALL students play cricket? [\[Comparing Quantities\]](#)

- (A) 12.5%
- (B) 65%
- (C) 10%
- (D) 100%
- (E) 15%
- (F) 20%
- (G) 8%

51. A baby's temperature is 2 deg C above the normal body temperature of 37 deg C. What does the thermometer read? [\[Heat\]](#)

- (A) 35 deg C
- (B) 39.5 deg C
- (C) 40 deg C
- (D) 38 deg C
- (E) 74 deg C
- (F) 39 deg C
- (G) 37 deg C

52. The length of a rectangle is 3 cm more than its width. If the perimeter is 38 cm, find the width. [\[Simple Equations\]](#)

- (A) 9.5 cm
- (B) 5 cm
- (C) 19 cm
- (D) 11 cm
- (E) 6 cm
- (F) 8 cm
- (G) 16 cm

53. A car travels at 54 km/h. How many metres does it cover in 1 second? [\[Motion & Time\]](#)

- (A) 54 m
- (B) 0.9 m
- (C) 20 m
- (D) 18 m
- (E) 15 m
- (F) 12 m

54. A trader marks goods 40% above cost and then gives a 10% discount. If the cost is Rs 500, what is the profit percent? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 40%
- (C) 20%
- (D) 26%
- (E) 30%
- (F) 10%
- (G) 36%

55. Water is heated steadily starting from 20 deg C. At what temperature does it normally begin to boil, and by how much must its temperature rise to reach that point? [\[Heat\]](#)
- (A) 100 deg C; a rise of 20 deg C
 - (B) 100 deg C; a rise of 100 deg C
 - (C) 100 deg C; a rise of 80 deg C
 - (D) 100 deg C; a rise of 120 deg C
 - (E) 0 deg C; a rise of 20 deg C
 - (F) 80 deg C; a rise of 60 deg C
56. Solve for x: $3x/5 = 12$. [\[Simple Equations\]](#)
- (A) 25
 - (B) 15
 - (C) 20
 - (D) 4
 - (E) 7.2
 - (F) 36
 - (G) 60
57. A train runs 200 km at 50 km/h, halts 20 minutes, then runs 150 km at 75 km/h. What is the total journey time? [\[Motion & Time\]](#)
- (A) 6 h 20 min
 - (B) 6 h 40 min
 - (C) 6 h 50 min
 - (D) 5 h 20 min
 - (E) 6 h
 - (F) 5 h
58. Out of 1500 votes, candidate A received 0.36 of them. How many votes did A get? [\[Comparing Quantities\]](#)
- (A) 54
 - (B) 415
 - (C) 540
 - (D) 36%
 - (E) 960
 - (F) 36
59. A 200 cm metal rod expands by 0.3 cm when heated. By how much would a 600 cm rod of the same metal expand under the same heating? [\[Heat\]](#)
- (A) 0.6 cm
 - (B) 3 cm
 - (C) 0.9 cm
 - (D) 0.1 cm
 - (E) 0.45 cm
 - (F) 0.3 cm

60. When a number is divided by 6 and then 5 is added, the result is 12. Find the number. [\[Simple Equations\]](#)

- (A) 2
- (B) 102
- (C) 7
- (D) 42
- (E) 36
- (F) 48
- (G) 17

61. A car drives 90 km at 30 km/h, then 120 km at 60 km/h. What is the average speed for the whole trip?

[\[Motion & Time\]](#)

- (A) 42 km/h
- (B) 21 km/h
- (C) 48 km/h
- (D) 45 km/h
- (E) 50 km/h
- (F) 40 km/h

62. Which is greater, and by how many percentage points: $\frac{3}{4}$ or 70%? [\[Comparing Quantities\]](#)

- (A) 70%, by 0.05 points
- (B) $\frac{3}{4}$, by 7.5 points
- (C) $\frac{3}{4}$, by 0.5 points
- (D) equal
- (E) $\frac{3}{4}$, by 5 points
- (F) 70%, by 5 points
- (G) $\frac{3}{4}$, by 25 points

63. Why is a room heater placed near the floor while an air-conditioner is fixed near the ceiling? [\[Heat\]](#)

- (A) Warm air rises and cool air sinks, so this sets up convection that spreads the temperature through the room
- (B) Electricity is safer near the ceiling
- (C) Cold air rises, so the AC must be high to catch it
- (D) Heat only travels downward, so heaters must be low
- (E) It is only for the look of the room
- (F) Warm air sinks and cool air rises, so the layout is reversed

64. Solve for x: $5(x - 2) = 3(x + 4)$. [\[Simple Equations\]](#)

- (A) 5.5
- (B) 1
- (C) 22
- (D) -11
- (E) 9
- (F) 11

65. Car A travels 240 km in 3 hours; Car B travels 270 km in 4.5 hours. Car A's speed is what percent of Car B's? [\[Motion & Time\]](#)

- (A) 80%
- (B) 133.3%
- (C) 120%
- (D) 125%
- (E) 75%
- (F) 60%

66. Find the amount due on Rs 12000 at 8% per annum simple interest after 2.5 years. [\[Comparing Quantities\]](#)

- (A) Rs 12960
- (B) Rs 14400
- (C) Rs 2400
- (D) Rs 13920
- (E) Rs 16800
- (F) Rs 9600
- (G) Rs 15600

67. A heater raises the temperature of some water by 4 deg C each minute. Starting at 18 deg C, after how many minutes will the water reach 70 deg C? [\[Heat\]](#)

- (A) 52 min
- (B) 26 min
- (C) 17.5 min
- (D) 18 min
- (E) 70 min
- (F) 12 min
- (G) 13 min

68. After spending one-third of his money, Aman had Rs 240 left. How much money did he start with?

[\[Simple Equations\]](#)

- (A) Rs 480
- (B) Rs 720
- (C) Rs 160
- (D) Rs 80
- (E) Rs 360
- (F) Rs 300

69. An athlete runs at 8 m/s. What is this speed in km/h? [\[Motion & Time\]](#)

- (A) 8 km/h
- (B) 40 km/h
- (C) 16 km/h
- (D) 2.22 km/h
- (E) 28.8 km/h
- (F) 80 km/h
- (G) 144 km/h

70. A shirt's price fell from Rs 800 to Rs 600. What is the percentage decrease? [\[Comparing Quantities\]](#)

- (A) 20%
- (B) 25%
- (C) 33.3%
- (D) 15%
- (E) 75%
- (F) 200%
- (G) 30%

71. A freezer is at -8 deg C and the kitchen is at 34 deg C. What is the temperature difference between them? [\[Heat\]](#)

- (A) 26 deg C
- (B) 42 deg C
- (C) -26 deg C
- (D) 50 deg C
- (E) -42 deg C
- (F) 30 deg C
- (G) 8 deg C

72. Solve for x: $2x - 3 = 5x - 18$. [\[Simple Equations\]](#)

- (A) 7
- (B) -5
- (C) 2
- (D) 6
- (E) 15
- (F) 3
- (G) 5

73. Two trains 360 km apart move toward each other at 60 km/h and 90 km/h. After how long do they meet? [\[Motion & Time\]](#)

- (A) 1 h 24 min
- (B) 4 h
- (C) 2 h 24 min
- (D) 2 h
- (E) 3 h
- (F) 2 h 40 min
- (G) 6 h

74. If 5 kg of apples cost Rs 400, what is the cost of 8 kg at the same rate? [\[Comparing Quantities\]](#)

- (A) Rs 600
- (B) Rs 500
- (C) Rs 720
- (D) Rs 640
- (E) Rs 800
- (F) Rs 256
- (G) Rs 320

75. In a hot desert, why do people often wear loose, white clothes? [\[Heat\]](#)

- (A) White absorbs heat and warms the body in the cold night
- (B) White reflects most of the heat radiation and loose cloth lets air circulate, keeping the body cooler
- (C) Tight dark clothes would reflect more sunlight
- (D) White is the only colour available in deserts
- (E) Loose clothes block all sweating
- (F) Dark colours keep the body coolest in strong sun

76. A father is three times as old as his son. In 12 years, he will be twice as old as his son. Find the son's present age. [\[Simple Equations\]](#)

- (A) 12 years
- (B) 36 years
- (C) 48 years
- (D) 24 years
- (E) 6 years
- (F) 4 years
- (G) 18 years

77. A delivery van travels 30 km in 1 h, 40 km in 0.5 h, and 50 km in 1.5 h. What is its average speed?

[\[Motion & Time\]](#)

- (A) 60 km/h
- (B) 120 km/h
- (C) 48 km/h
- (D) 45 km/h
- (E) 40 km/h
- (F) 35 km/h
- (G) 53.3 km/h

78. Ravi's salary is 25% more than Sita's. By what percent is Sita's salary less than Ravi's? [\[Comparing](#)

[Quantities\]](#)

- (A) 12.5%
- (B) 16.7%
- (C) 33.3%
- (D) 40%
- (E) 25%
- (F) 20%
- (G) 75%

79. A metal block is heated and its temperature goes 20 deg C -> 24 deg C -> 30 deg C -> 38 deg C over three equal time steps. In which step was the rise largest, and by how much? [\[Heat\]](#)

- (A) Second step, by 6 deg C
- (B) Second step, by 10 deg C
- (C) Third step, by 18 deg C
- (D) First step, by 4 deg C
- (E) Third step, by 38 deg C
- (F) Third step, by 8 deg C

80. Solve for x : $x/2 - x/3 = 4$. [\[Simple Equations\]](#)

- (A) 48
- (B) 4
- (C) 12
- (D) 24
- (E) 8
- (F) 2
- (G) 6

81. A clock's pendulum has a time period of 0.5 s. How many complete oscillations does it make in 5 minutes? [\[Motion & Time\]](#)

- (A) 150
- (B) 500
- (C) 600
- (D) 2.5
- (E) 300
- (F) 1200
- (G) 60

82. A 50% discount followed by a 20% discount is equal to a single discount of what percent? [\[Comparing Quantities\]](#)

- (A) 70%
- (B) 50%
- (C) 40%
- (D) 55%
- (E) 60%
- (F) 30%
- (G) 65%

83. At night, the land cools faster than the sea. Which way does the breeze blow, and what is this breeze called? [\[Heat\]](#)

- (A) From sea to land; a land breeze
- (B) From land to sea; a monsoon breeze
- (C) From land to sea; a sea breeze
- (D) From sea to land; a sea breeze
- (E) From land to sea; it is a land breeze
- (F) No breeze blows at night

84. The sum of two numbers is 45, and one number is twice the other. Find the larger number. [\[Simple Equations\]](#)

- (A) 45
- (B) 35
- (C) 25
- (D) 20
- (E) 15
- (F) 30

85. A drone flies for 45 minutes at a steady 12 km/h. What distance does it cover? [\[Motion & Time\]](#)

- (A) 6.75 km
- (B) 8 km
- (C) 540 km
- (D) 10 km
- (E) 12 km
- (F) 18 km
- (G) 9 km

86. In how many years will Rs 6000 earn Rs 1440 as simple interest at 6% per annum? [\[Comparing Quantities\]](#)

- (A) 3 years
- (B) 6 years
- (C) 2 years
- (D) 5 years
- (E) 4.8 years
- (F) 4 years
- (G) 2.4 years

87. A laboratory thermometer is marked from -10 deg C to 110 deg C. How many degrees does its full scale cover? [\[Heat\]](#)

- (A) 10 deg C
- (B) 110 deg C
- (C) 120 deg C
- (D) 220 deg C
- (E) 90 deg C
- (F) 130 deg C
- (G) 100 deg C

88. Solve for x: $6x + 11 = 4x + 25$. [\[Simple Equations\]](#)

- (A) 14
- (B) 18
- (C) 5
- (D) 6
- (E) 8
- (F) 7
- (G) 3.6

89. A 180 km journey must be finished in 2 hours 15 minutes. What steady speed is required? [\[Motion & Time\]](#)

- (A) 90 km/h
- (B) 80 km/h
- (C) 85 km/h
- (D) 100 km/h
- (E) 72 km/h
- (F) 60 km/h
- (G) 75 km/h

90. A 40-litre mixture is 30% acid. How many litres of pure acid must be added to make it 50% acid?

[Comparing Quantities]

- (A) 10 litres
- (B) 20 litres
- (C) 4 litres
- (D) 16 litres
- (E) 12 litres
- (F) 8 litres
- (G) 24 litres

91. Tea A cools from 85 deg C to 55 deg C; tea B cools from 70 deg C to 46 deg C. Which cooled more, and by how much more? [Heat]

- (A) Tea A, by 6 deg C
- (B) Tea B, by 6 deg C
- (C) Equal cooling
- (D) Tea A, by 9 deg C
- (E) Tea B, by 24 deg C
- (F) Tea A, by 54 deg C

92. Two angles of a triangle are equal, and the third is 30 deg more than each of them. Find each of the equal angles. [Simple Equations]

- (A) 60 deg
- (B) 75 deg
- (C) 50 deg
- (D) 30 deg
- (E) 80 deg
- (F) 45 deg

93. A car travels at 25 m/s for 12 minutes. How far does it travel, in km? [Motion & Time]

- (A) 15 km
- (B) 18000 km
- (C) 36 km
- (D) 300 km
- (E) 18 km
- (F) 30 km
- (G) 1.8 km

94. A restaurant bill is Rs 2000. A 5% tax is added, then a 10% service charge is added on the new total. What is the final amount? [Comparing Quantities]

- (A) Rs 2400
- (B) Rs 2200
- (C) Rs 2310
- (D) Rs 2315
- (E) Rs 2300
- (F) Rs 2150
- (G) Rs 2100

95. Why are the handles of cooking pots often made of plastic or wood? [\[Heat\]](#)

- (A) They are good conductors but cool down instantly
- (B) They are good conductors, so heat passes through quickly
- (C) They are cheaper than any metal
- (D) They reflect heat back into the pot
- (E) They are poor conductors of heat, so they stay cool enough to hold safely
- (F) They absorb all the heat from the food

96. Solve for x: $8 - 3(x - 2) = 5$. [\[Simple Equations\]](#)

- (A) 3
- (B) 4
- (C) 2
- (D) 9
- (E) -3
- (F) 1

97. A car covers two equal distances, one at 20 km/h and the other at 30 km/h. What is its average speed for the whole trip? [\[Motion & Time\]](#)

- (A) 25 km/h
- (B) 22 km/h
- (C) 12 km/h
- (D) 50 km/h
- (E) 24 km/h
- (F) 26 km/h
- (G) 28 km/h

98. A recipe for 4 people needs 300 g of flour. How much flour is needed for 10 people at the same rate?

[\[Comparing Quantities\]](#)

- (A) 1200 g
- (B) 700 g
- (C) 750 g
- (D) 120 g
- (E) 450 g
- (F) 600 g

99. Over 5 days the midday temperature averaged 36 deg C. The first four days averaged 34 deg C. By how many degrees was the fifth day ABOVE the 5-day average? [\[Heat\]](#)

- (A) 8 deg C
- (B) 4 deg C
- (C) 44 deg C
- (D) 10 deg C
- (E) 2 deg C
- (F) 34 deg C

100. The average of a number and 18 is 25. Find the number. [\[Simple Equations\]](#)

- (A) 28
- (B) 50
- (C) 32
- (D) 14
- (E) 43
- (F) 36
- (G) 21.5